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Curriculum

for the Master's degree programme in
Artificial Intelligence and Cybersecurity

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Curriculum for the Master's degree programme in

Artificial Intelligence and Cybersecurity

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Section 1 General remarks

- (1) The Master's degree programme in *Artificial Intelligence and Cybersecurity* comprises 120 European Credit Transfer System Credits (ECTS credits). This corresponds to an anticipated study duration of four semesters. The Master's degree programme in *Artificial Intelligence and Cybersecurity* is assigned to the group Studies in Engineering Sciences pursuant to Section 54 (1) of the Austrian Universities Act 2002 (UG).
- (2) The academic workload for the individual is indicated in ECTS credits; the workload for one year should amount to 1,500 full hours, for which 60 ECTS credits are awarded (Section 54 (2) UG). The workload comprises independent study as well as the semester hours/contact hours, including participation in assessment procedures.
- (3) The Master's degree programme in *Artificial Intelligence and Cybersecurity* is offered in English as a joint programme of study in accordance with Section 54d UG in the form of a double degree programme in cooperation with the University of Udine. Students seeking to complete the double degree programme must obtain 30 ECTS credits at the University of Udine.
- (4) The Master's degree programme is taught in English.

Section 2 Qualification profile and competences

The qualification profile describes the academic and professional skills that students gain by completing the degree programme.

- (1) **Initial situation:** The Master's degree programme in *Artificial Intelligence and Cybersecurity* at the University of Klagenfurt focuses on (linking) the areas of artificial intelligence and cybersecurity. The aim is to enable graduates to use artificial intelligence and cybersecurity solutions in a high-quality, ethical manner. This requires sound basic knowledge and skills in the field of informatics/mathematics and deeper knowledge and skills in artificial intelligence and cybersecurity as well as relevant ethical aspects.
Artificial intelligence and cybersecurity are, both individually and combined, dynamic areas of knowledge that are subject to rapid change and pursue constant growth. In practice, both areas of knowledge are typically used as "foundational and enabling technologies", which facilitate other applications in the first place. Graduates of the Master's programme must therefore be able to expand their knowledge independently and work effectively in teams. Consequently, methodological, social and personal skills are necessary for the successful realisation, introduction, and beneficial utilisation of artificial intelligence and cybersecurity solutions.
A combination of independent as well as team-based research and development together with the strategic use of classical forms of teaching are intended to create ideal conditions for developing these skills.
- (2) **Intended learning outcomes and qualifications:** Building on the initial situation, there are three main fields of competence in which students completing the Master's degree programme in *Artificial Intelligence and Cybersecurity* gain knowledge and skills:

1) Specialised skills in artificial intelligence and cybersecurity, taking into account their ethical use; 2) individual and team skills; 3) practical scientific research and development. The knowledge and skills to be gained are described below in the form of learning outcomes for each field of competence:

- *Specialised skills* in artificial intelligence and cybersecurity: Knowledge and skills in the fields are acquired through appropriate basic and specialist subjects in artificial intelligence and cybersecurity. The teaching is based on current research results and thus conveys the current state of the art to students. The aim of the courses offered is to prepare students for practical, scientifically sound work and to teach them to conduct technology impact assessments. Social, ethical and legal aspects are both embedded in individual courses and addressed in specific courses.
- *Individual and team skills*: The knowledge and skills gained through the courses enable students to research information independently as well as in teams and to develop solutions taking into account social, ethical and legal aspects. They also enable students to critically examine scientific findings, technical solutions and their impact on society.
- *Research experience and problem-solving competence (including scientific work)*: Students have the opportunity to work on a project concerning a complex issue. This gives them a feel for and experience of industrial or research projects. In this way, graduates boost their skills for solving a wide range of problems in business or administration, or when addressing research questions in the academic sphere.

- (3) **Professional fields and fields of activity**: Graduates of the Master's degree programme in *Artificial Intelligence and Cybersecurity* possess special expertise in the fields of artificial intelligence and cybersecurity. They are able to develop and implement complex solutions based on artificial intelligence and/or cybersecurity, taking into account social, ethical and legal aspects. The Master's degree programme also prepares graduates for a doctoral programme in the field of Informatics.

Section 3 Admission requirements

- (1) Admission to a Master's degree programme is contingent upon completion of a relevant Bachelor's degree programme, another relevant degree programme of at least the same higher education level completed at a recognised domestic or foreign post-secondary educational institution, or a degree programme as defined in the curriculum of the Master's degree programme (Section 64 (3) UG).
- (2) The Bachelor's degree programmes in Applied Informatics, Informatics, Information Management, Management Information Systems, Information and Communications Engineering, Robotics & Artificial Intelligence, and the teacher training programme in Computer Science and Digital Education at the University of Klagenfurt as well as the Corso di Laurea in Informatica at the University of Udine are classed as relevant in any case.
- (3) In order to compensate for substantial differences in terms of the content of other relevant programmes of at least the same level of higher education completed at a recognised domestic or foreign post-secondary educational institution, supplementary

examinations may be prescribed, which must be completed by the end of the second semester of the Master's programme. The Rectorate may determine which of these supplementary examinations are prerequisites for sitting the examinations stipulated in the curriculum of the Master's degree programme (Section 64 (3) UG).

- (4) The following qualitative admission conditions are stipulated for the Master's degree programme:

Students who have completed a Bachelor's degree programme in engineering sciences or a Bachelor's degree programme in mathematics or statistics or an equivalent degree programme at another domestic or international university, university of applied sciences or other recognised post-secondary educational institution not covered by Para. 1, shall be admitted to the Master's degree programme if they provide evidence of knowledge in core areas of informatics (algorithms and data structures, discrete mathematics, basic probability theory and statistics, at least basic programming experience).

- (5) Persons whose first language is not English must be proficient in the English language at level B2 of the Common European Framework of Reference for Languages (CEFR).

Section 4 Academic degree

- (1) Graduates of this Master's degree programme will be awarded the academic title "Master" accompanied by the words "of Science" (abbreviated to "MSc"). If used, this title must follow after the name.
- (2) Graduates who obtain at least 30 ECTS credits in the double degree programme at the University of Udine will be issued a joint certificate with the University of Udine in accordance with Section 87 (5) UG as well as being awarded the academic degree.

Section 5 Structure & organisation of the degree programme/intended learning outcomes

The required subjects, the guided electives and the open electives must be completed as part of the Master's degree programme in *Artificial Intelligence and Cybersecurity*. In addition, a Master's thesis must be prepared, the associated exclusive tutorial and critical reflection must be completed. Table 1 shows the structure of the Master's degree programme in *Artificial Intelligence and Cybersecurity*.

Table 1: The structure of the Master's degree programme in *Artificial Intelligence and Cybersecurity*

Subject/ academic achievement	Designation	Intended learning outcomes	ECTS credits
Required subjects	1 Foundations of Artificial Intelligence and Cybersecurity	Students are able to identify the essential technical, ethical and legal bases for using artificial intelligence and cybersecurity, and can: - Identify and apply technical and mathematical principles relevant to	12

		artificial intelligence and cybersecurity - Explain the social, ethical, and legal principles relevant to artificial intelligence and cybersecurity	
	2 Artificial Intelligence	Students are able to explain the key concepts and methods of artificial intelligence, and can: - Design artificial intelligence systems - Identify and apply appropriate methods for specific areas of interest in the field of artificial intelligence - Analyse the limits of artificial intelligence systems - Identify and analyse social and ethical aspects in the specific design of artificial intelligence systems	14
	3 Cybersecurity	Students are able to explain the key concepts and methods of cybersecurity, and can: - Define and analyse cybersecurity goals - Select and implement cryptographic methods appropriately - Examine implementations for accuracy - Identify and analyse social and ethical aspects in the specific design of a cybersecurity system	8
Guided electives - choice of a specialist subject	4 Specialisation in Artificial Intelligence and Cybersecurity	Students are able to identify advanced concepts and methods of artificial intelligence and/or cybersecurity and can implement them in a practical manner in relevant fields of application.	42
	5 Responsible Innovation	Students are able to identify the principles and methods of responsible innovation, in particular gender aspects, and can apply them in the area of artificial intelligence and cybersecurity.	8
Open electives	6 Open electives	Students can choose from the full range of courses offered by the Universities of Klagenfurt and Udine and acquire additional skills accordingly.	6

Master's thesis	7 Master's thesis including exclusive tutorial and Master's seminar	Students are able to study complex questions in the area of artificial intelligence and/or cybersecurity according to the state of the art and to implement correct and ethical solutions.	23+2+2
Final overall examination	8 Final overall examination		3
		Total:	120

Section 6 Study-related period abroad/mobility

- (1) It is highly recommended for all students of the Master's degree programme to complete a study-related period abroad within the framework of their degree programme. Transnational EU, state or university mobility programmes can be used for this purpose. Examinations and other academic achievements completed during a study-related period abroad will be recognised in accordance with the provisions of Section 78 UG for the examinations prescribed in the curriculum. The second or third semester is recommended as the mobility window.
- (2) At the request of regular students who wish to carry out parts of their studies abroad, a preliminary decision shall be issued in advance to determine which of the proposed examinations and other academic achievements can be recognised (Section 78 (5) UG). In any case, interested students are advised to contact the respective Programme Director in advance to discuss possible and intended recognition.

Section 7 Required subjects

- (1) Required subjects are subjects significant to the degree programme and for which examinations must be taken. The courses for the required subjects can be found in Table 2 below. Students must complete a total of 34 ECTS credits in the required subjects.
- (2) If the course content of the required subjects 1.1, 2.1 and/or 3.1 has demonstrably already been completed as part of another degree programme, subjects from corresponding guided electives comprising at least the same number of ECTS credits must be selected in consultation with the Programme Director.
- (3) The courses for the required subjects can be found in Table 2 below:

Table 2: Required subjects and assigned courses

	Course title	Course type	ECTS credits
Foundations of Artificial Intelligence and Cybersecurity	1.1 Fundamentals of Artificial Intelligence and Cybersecurity	VI	4
	1.2 Social, Ethical and Legal Aspects of Artificial Intelligence and Cybersecurity	VI	4
	1.3 Scientific Writing	VI	4
	Total:		12
Artificial Intelligence	2.1 Introduction to Artificial Intelligence 1	VI	3
	2.2 Knowledge Representation, Reasoning and Uncertainty	VI	5
	2.3 Machine Learning and Deep Learning	VI	6
	Total:		14
Cybersecurity	3.1 Introduction to Cybersecurity	VI	8
	Total:		8

Section 8 Guided electives

- (1) Guided electives are those subjects that students choose according to the provisions of the curriculum. A total of 50 ECTS credits must be completed in guided electives from the areas of Specialisation in Artificial Intelligence and Cybersecurity and Responsible Innovation. Of these, 42 ECTS credits must be selected from the area of Specialisation in Artificial Intelligence and Cybersecurity and 8 ECTS credits from the area of Responsible Innovation.
- (2) The courses, which are assigned to the guided electives, serve to deepen the knowledge, methods and skills in the (application) fields of Artificial Intelligence and Cybersecurity as well as Responsible Innovation. This includes technical applications as well as mathematical subjects.
- (3) Projects, hereinafter referred to as "Small Project" and "Large Project" according to Table 3, items 4.5 and 4.6, are used for practical or research-oriented application or extension of the acquired knowledge and skills in the fields of artificial intelligence and cybersecurity. One of the two projects (4.5 or 4.6) can be completed as part of the guided elective Specialisation in Artificial Intelligence and Cybersecurity. Students can choose the subject area or topic of the project or choose from a number of suggestions. Before beginning the project, approval must be obtained from the supervisor.
- (4) A written report to document the content, results and experiences must be drawn up as part of the project. The supervising university lecturer evaluates the project on the basis of this report and a discussion.
- (5) The courses assigned to the guided electives can be found in Table 3 below:

Table 3: Guided electives and assigned courses (LVs)

	Course title	Course type	ECTS credits
Specialisation in Artificial Intelligence and Cybersecurity	4.1 Advanced Topics in Artificial Intelligence 1	VI	6
	4.2 Advanced Topics in Artificial Intelligence 2	VI	6
	4.3 Advanced Topics in Cybersecurity 1	VI	6
	4.4 Advanced Topics in Cybersecurity 2	VI	6
	4.5 Small Project in Artificial Intelligence and Cybersecurity	KS	10
	4.6 Large Project in Artificial Intelligence and Cybersecurity	KS	20
	4.7 Relevant courses	*1	*2
		Total:	42
Responsible Innovation	5.1 Communicating Artificial Intelligence and Cybersecurity Research	SE	2
	5.2. Responsible Engineering of Artificial Intelligence and Cybersecurity Systems	SE	6
	5.3. Gender and Technology	*1	*2
	5.4 Relevant courses	*1	*2
		Total:	8

Section 9 Open electives

- (1) Open electives are those subjects that students can freely choose from the range of courses offered by recognised domestic and foreign universities. Courses completed as a prerequisite to study or to gain general or special eligibility for university admission cannot be used for the open electives.
- (2) Students are required to complete 6 ECTS credits in open electives.
- (3) In the case of courses that have been completed at other recognised domestic or international post-secondary educational institutions, the responsible Programme Director will decide whether recognition as an open elective for the chosen programme of study makes sense academically or with regard to professional activities.

Section 10 Courses with a limited number of participants

- (1) The maximum number of participants permitted on each of the following types of course is as follows:
 - Integrated lecture (VI): 30, with the exception of the "Social, Ethical and Legal Aspects of Artificial Intelligence and Cybersecurity" course, with a maximum of 20 participants;

¹ Different types of courses are possible.

² Different ECTS credits are possible.

- Seminar (SE): 15, with the exception of the "Responsibility in Artificial Intelligence & Cybersecurity: Reflecting the social and ethical dimensions in the Master's thesis" course with a maximum of 10 participants.
- (2) If the number of applications for these courses exceeds the number of places available, students will be accepted in accordance with the following procedure:
 - (a) Students for whom the course is a required subject or a guided elective according to their curriculum are given priority.
 - (b) If the number of applications still exceeds the number of available places, students are ranked based on the already acquired ECTS credits for the curriculum that stipulates the course in question as a required subject/guided elective. A higher total is ranked higher.
 - (c) In addition, parallel courses are to be offered if necessary.
 - (3) For courses drawn from other curricula, the maximum numbers found in the respective curricula apply.
 - (4) Depending on the didactic requirements and the availability of spatial, budgetary and other resources, the Programme Director may determine maximum numbers of participants that deviate from these stipulations.

Section 11 Master's thesis

- (1) The Master's thesis is the academic thesis that demonstrates the student's ability to achieve adequate standards of content and methodology when independently addressing scholarly topics. The assignment for the Master's thesis should be chosen in such a way that it is possible and reasonable to expect a student to complete it within six months. Subject to approval by the supervisor, the Master's thesis may be written in a language other than German. A number of students may jointly address a topic, provided that the performance of individual students can be assessed.
- (2) The subject of the Master's thesis must be chosen from the required subjects (with the exception of the subject "Foundations of Artificial Intelligence and Cybersecurity") or from the guided electives (with the exception of the subject "Responsible Innovation").
- (3) The Master's thesis comprises 23 ECTS credits. Accompanying the Master's thesis, the exclusive tutorial (2 ECTS credits) offered by the Master's thesis supervisor must be completed along with a reflection within the framework of the Master's seminar "Responsibility in Artificial Intelligence & Cybersecurity: Reflecting the social and ethical dimensions in the master thesis" (2 ECTS credits).
- (4) The reflection drawn up within the framework of the Master's seminar in accordance with paragraph 3 shall be integrated into the Master's thesis. The seminar must be successfully completed before submission of the Master's thesis.
- (5) Pursuant to Part B, Section 18, Para. 4 and 2a of the University Statutes, both the topic and the Master's thesis supervisor must be approved by the Rector of Studies.

The application must be made prior to starting work. The student is permitted to change supervisor up until the time when the Master's thesis is submitted. Supervision by two persons who are authorised to supervise is permissible on a case-by-case basis, where there is good reason to do so (interdisciplinary focus of the topic).

- (6) The completed Master's thesis must be submitted to the Rector of Studies in electronic format. Upon the request of the supervisor, the author must provide them with a bound copy of the thesis. The supervisor will have two months from the date of submission to assess the Master's thesis.

Section 12 Examination regulations

- (1) Course examinations for lectures (VO) are - preferably in written form - held at the end of or after the course as a single examination and cover the course material. The purpose is to assess how successfully students have participated in the course and to demonstrate the student's mastery of the knowledge, methods and skills gained in the course. In particular, the educational objectives defined in the qualification profile (Section 2) must be used as a benchmark.
- (2) All other types of courses are subject to ongoing assessment; attendance is compulsory. Internships/projects (PR) are assessed through accompanying checks and also through written and oral examinations as well as on the basis of the success of practical activities. Students' written and oral contributions (especially seminar papers, presentations in seminars and participation in discussions) are used as a benchmark for assessment in seminars (SE) and exclusive tutorials (PV). In interactive lectures (VI), the mode of examination is to be determined based on the character of the course and the educational objectives.
- (3) The course instructors must provide information in a suitable manner about the contents, teaching methods and learning outcomes of their courses and the contents, methods, criteria of course assessment and examinations in the electronic course registration system (ZEUS) prior to the start of each semester (Section 76 (2) UG).
- (4) Examinations that have already been used for the completion of studies regarded as admission requirements cannot be used again to complete the programme of studies in the Master's degree programme.
- (5) To graduate from the Master's degree programme in *Artificial Intelligence and Cybersecurity*, students are required to successfully complete the following course components:
 - (a) The courses for the required subjects, both guided electives and the open electives;
 - (b) The Master's thesis, the associated exclusive tutorial as well as the seminar in accordance with Section 11 (3);
 - (c) The final overall oral examination held before an examination board according to Para. 6.

- (6) The prerequisite for applying for the final overall examination held before an examination board is the successful completion of the requirements listed under Para. 5, items (a) - (b). The final overall oral examination takes place in front of an examination board comprising at least three persons. In the case of the double degree programme, a representative from the University of Udine is a member of the examination board. The examination includes:
 - (a) The subject that the topic of the Master's thesis falls under (in the form of a presentation and defence of the Master's thesis); 1 ECTS credit
 - (b) The guided elective Responsible Innovation; 1 ECTS credit
 - (c) Another guided elective that is different to lit. a) and b); 1 ECTS credit.
- (7) The regulations of the Statutes of the University of Klagenfurt, Part B and the Universities Act as amended apply to the implementation and re-taking of examinations.

Section 13 Effective validity

- (1) This curriculum will enter into force on 1 October 2020 following announcement in the University of Klagenfurt university bulletin and will apply to all students who commence their Master's degree programme from the 2020/21 winter semester onwards.
- (2) The non-structural amendments to the curriculum, as announced in the university bulletin of the University of Klagenfurt dated 7 May 2025, issue 15, no. 92.1, will enter into effect on 1 October 2025. All students enrolled in the Master's programme will be subject to the amended curriculum upon its entry into effect.

Section 14 Transitional provisions

Students who have already commenced work on both projects in accordance with Section 8 (3) when this version 20W.2 comes into effect are entitled to use these projects to complete their degree. Students who have commenced their studies prior to the effective date of this version are entitled to use other guided electives for the completion of their studies in place of the required subject 1.3 pursuant to Section 7 (3) (Scientific Writing).

APPENDIX Non-binding recommended course of study

The following tables suggest how the ECTS credits could be distributed across the semesters.

Table 4: Non-binding recommended course of study

	1st semester	2nd semester*	3rd semester*	4th semester
1.1 Fundamentals of Artificial Intelligence and Cybersecurity	4 ECTS credits			
1.2 Social, Ethical and Legal Aspects of Artificial Intelligence and Cybersecurity		4 ECTS credits		
1.3 Scientific Writing	4 ECTS credits			
2.1 Introduction to Artificial Intelligence 1	3 ECTS credits			
2.2 Knowledge Representation, Reasoning and Uncertainty	5 ECTS credits			
2.3 Machine Learning and Deep Learning	6 ECTS credits			
3.1 Introduction to Cybersecurity		8 ECTS credits		
5.1 Communicating Artificial Intelligence and Cybersecurity Research		2 ECTS credits		
5.2. Responsible Engineering of Artificial Intelligence and Cybersecurity Systems		6 ECTS credits		
Choice of courses from guided and open electives	8 ECTS credits	10 ECTS credits		
Choice of courses from guided and open electives/stay at partner university			30 ECTS credits	
Responsibility in Artificial Intelligence & Cybersecurity: Reflecting the social and ethical dimensions in the Master's thesis				2 ECTS credits
Master's thesis (incl. exclusive tutorial)				25 ECTS credits
Final overall examination				3 ECTS credits
Total:	30 ECTS credits	30 ECTS credits	30 ECTS credits	30 ECTS credits

* The second or third semester is recommended as the mobility window within the meaning of Section 6 'Study-related stay abroad/mobility'.